

■ 2.5.4 Program Flow

When you use a procedure in *Mathematica*, its elements are usually executed one after another. In some cases, you may want to modify the order of execution. You may for example want to stop executing the elements, and just return the result you have. There are several functions that you can use to make these kinds of changes in the “flow” of your program.

<code>Return[expr]</code>	return the value <i>expr</i> , exiting all procedures in a function
<code>Break[]</code>	exit the nearest enclosing <code>While</code> or <code>For</code>
<code>Continue[]</code>	go to the next step in <code>While</code> or <code>For</code>
<code>Goto[name]</code>	go to the element <code>Label[name]</code> in the current procedure
<code>Throw[expr]</code>	return <i>expr</i> as the value of nearest enclosing <code>Catch</code> (non-local return)

Basic control flow constructs.

This defines a function which returns 0 if it finds that `t == 0`. You could just write `t` instead of `Return[t]` at the end of the procedure.

```
In[1]:= f[y_, x_] :=
      Block[ {t}, t = D[y, x]; If[t==0, Return[zero]];
      t = t^2 + 1; Return[t] ]
```

`t` is found to be 0, so `zero` is returned.

```
In[2]:= f[4, x]
Out[2]= zero
```

Now `t` is nonzero, so the rest of the procedure is executed.

```
In[3]:= f[x^2, x]
Out[3]= 1 + 4 x^2
```

This function “throws” `overflow` if its argument is greater than 10.

```
In[4]:= g[x_] := If[x > 10, Throw[overflow], x!]
```

`overflow` is not generated.

```
In[5]:= 2 + g[5]
Out[5]= 122
```

`g[20]` throws `overflow`, which is returned as the value of the enclosing `Catch`.

```
In[6]:= Catch[ 2 + g[20] ]
Out[6]= overflow
```

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Now the `Catch` has no effect.

```
In[7]:= Catch[ 2 + g[5] ]  
Out[7]= overflow
```

If you are familiar with the C programming language, you will notice that `Return`, `Break` and `Continue` work essentially the same way in *Mathematica* as in C.